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Commentary

Geographies of experiment

"Science itself invites us to read the scene of experimentation, its fractured promises and articulated procedures, the historical renewals, stalls or question marks, which the experimental disposition has generated."

Avital Ronell (2003b, page 656)

"Experimentation has a life of its own."

Ian Hacking (1983, page 150)

The vagaries of experiments, as Ian Hacking argued over two decades ago, "have been neglected for too long" (Hacking, 1983, page xv). If geography has been rather lethargic in investigating the legacy of experiment for its own practices, the vocabulary of experiment has never been more common in geographical discourse—ranging through discussions of science and biopolitics as well as capital and modernity. We offer here a provocation for geographers to take seriously experiment through attention to experimental spaces and their varied *polities*.

Geographies of scientific experiment

Although Hacking's charge was directed explicitly at philosophers of science, sociologists and anthropologists have taken up his historicist imperative by producing narratives tracing the relations between matters of experiment and their social lives (Collins, 1992; Galison, 1997; Gooding et al 1989a; 1989b; Rheinberger, 1997; Shapin and Schaffer, 1985; Voskuhl, 1997). As such studies of experimentation have gained prominence, cultural and historical geographers have belatedly drawn attention to the spatialities that embed, and are produced through, scientific practices (Bingham, 1996; Driver, 1994; Livingstone, 1995; 2002; 2003; Murdoch, 1998; Naylor, 2005; Whatmore, 2002; Withers, 2001).

As geographers of science, we applaud recent conversations between historians of science and historical geographers around the issue of how the spaces of scientific activity might be best conceptualized and addressed (Livingstone, 2005; Shapin, 2003). In such discussions, however, the vivacity of scientific conduct has sometimes been lost in arguments over the potential "category mistake" of the spatiality of science (Shapin, 2003, page 90). Mirroring whispers heard many times in debates across histories and philosophies of geographical practice, critics of attempts to institutionalize geographies of science have charged that "the spatiality of science is not a 'factor'" like class, religion, or nationality (Shapin, 2003, page 90). Rather, the spatialities of science, as necessary conditions for its constitution, are much more than that. Indeed, as such malcontents bemoan, what could an *a*spatial science look like?

Perhaps too much energy has been invested in these debates. As a result, we call for discussions of a range of experimental contexts drawn from the physical, environmental, and physiological sciences that attempt to move forward the agenda for a fuller historical geography of science (Lorimer and Spedding, 2005). We urge geographers of science to deploy ethnographic and historical insight in the development of a geographical sensitivity to the spaces of actual experiments. In so doing, we hope to bring the effervescence of scientific experiment to light. This involves attention to the full range of bodies, texts, and practices that constitute spaces of experimentation. We would wish to capture the nonverbal and gestural dimensions of experimental activity (Schaffer, 1997; Sibum, 1995), although we are not calling for the "reperformance of an experiment" along the lines that Adelheid Voskuhl (1997, page 337) has undertaken with careful replicas of John Herschel's 'actinometer'.[(1)] At the expense of efforts at disciplinary institutionalization, therefore, we would rather advocate that geographers of science focus instead on the presentation of those "wonderfully intriguing, vivid and original stories" which, in weaving together bodies, practices, and objects, have been so attractive to all students of the social and cultural particularities of scientific activities (Shapin, 2003, page 90).

We would stress, moreover, that this is not simply a call for more empirical studies of experiments. Rather, it is through a renewed focus on the actualities of experimentation that we want to reinvigorate the *politics* of experiment.

The politics of experiment

The terminology of experiment has proved seductive in recent discussions in geography. Experiment has been used to conceive of the political economy of capitalism (Thrift, 2001). "[C]osmopolitical experiments" have been written which attempt to produce new ways of debating the political collective while offering an alternative to the old settlements of 'Science' and 'Politics' (Hinchliffe et al, 2005, page 644). And, of course, it is in the biopolitical experiment of the Camp that we have seen the emergence of the carceral laboratories so crucial to the spatialization of global politics in a 'New American Century' (Minca, 2005; see also Agamben, 2005).

These efforts are to be applauded, but we find another impulse in the substance of experiment. We urge caution against the adoption of a blithe experimentalism or the casual use of a slack metaphorics. Experimental geographies, it must be remembered, are not necessarily geographies of experiment. In noting this, we want not only to emphasize an ongoing engagement with the scene of experimentation—with its material cultures, epistemic protocols, and textual products—but also to heed Ronell's urgent call for further critical reflection on the singularity of the modern experimental turn (Ronell, 2003a; 2003b; 2005). Rather less than definitional pedantry, this returns us squarely to the issue of the social ordering inculcated in the 'experimental life' (see Shapin and Schaffer, 1985).

As well as investigating what experiments might mean to scientists, we thus wish to shift attention to the production of cultural and epistemic geographies of experimentation, focusing on how such practices circulate through and extend into putatively nonscientific zones. What we are after here might include recent research by geographers on the unseemly affinities between experimental matters of fact, biopolitical modes of rationality, and historically specific forms of governance (Castree, 2003; Gagen, 2004; Nash, 2005; Parry, 2004, 2005).

These particularities of experiment, we argue, are intertwined with wider geographies that move beyond laboratories and field camps (Powell, 2007; Vasudevan, 2007). Indeed, for Ronell, what we have come to routinely understand as 'experiment' or 'test' ought to be read as one of the prevailing figures of our modernity. "Today's world", she writes, "is ruled conceptually by the primacy of testing—nuclear, drug, HIV, admissions, employment, pregnancy, and DNA tests, the SAT, GRE, and MSAT" (Ronell, 2003a, page 563). Put boldly, the test has restructured the field of everyday life. This is, we realize, a provocative claim, and one whose significance merits closer scrutiny. After all, as Ronell points out, "what is it that links the recondite behaviors of lab culture to drug experimentation, experimental theater, thought experiments, [and] political acts?" (Ronell, 2003b, page 657). What, to put it another way, is the *nature* of experimental enquiry?

(1) An 'actinometer' was an early-19th-century instrument that measured the intensity of solar radiation (Voskuhl, 1997).

These questions are substantively political. Torture and experiment, after all, have been expressly linked since the investigations of Francis Bacon (Ronell, 2005). Nietzsche "foresaw the concentration camp as the most unrestricted experimental laboratory in modern history" (Ronell, 2005, pages 6-7). Ronell, in turn, sees experimental testing coming to require an "*unprecedented colloquy of witnesses, evidence, repeatability, and an entire network of apparatuses*" (2005, page 19, original emphasis). The modern test depends on a constitutive spatiality.

If, in the case of Ronell, a response to these increasingly insistent imperatives—both epistemologically and ontologically—would involve a closer look at a host of phenomena "that appear to have flown beneath philosophical radars" (2003b, page 657), the discipline of geography has only begun the task of responding to "the question of testing" (Ronell, 2003b, page 659). With this in mind, we ultimately wish to encourage authors to present case studies of scientific practice in specifically experimental settings, focusing not only on the *where* of experimentation, but also on the mechanisms which adjudicate *what* eventually counts as proper experimental knowledge. In attempts to move forward the agenda for a fuller historical geography of science, we believe that the question is less one of a pro forma reading of the role that space plays in the production and circulation of scientific meaning, than it is to articulate spatial histories of experimentation—"sited histories" to borrow Peter Galison's (1997, page 63) felicitous phrase—that fully acknowledge the rich, varied, and often irremediably local circumstances through which experimental matters of fact are produced, contested, and institutionalized.

We are painfully aware that we have only been able here to gesture towards the forms that such geographies of experiment might take. But we are of the firm conviction that it is only through geographical focus on actual sites of experimentation that such studies might take on lives of their own. In these testing times, we see a pressing need for investigation of the provenance of the experiment.

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